SKILL REFERENCE

Website optimizer

The complete contents of the website-optimizer skill in one document. It combines SKILL.md, references/checklist.md and references/values.md, with the original text kept in full.

Contents

Part 1: Website optimizer (SKILL.md) . . . 2

Workflow . . . 2

The short version of the tells . . . 4

Honesty rule . . . 4

Reference files . . . 4

Part 2: Review checklist (references/checklist.md) . . . 5

1. Accessibility . . . 5

2. Layout and spacing . . . 6

3. Typography . . . 6

4. Colour and surfaces . . . 7

5. Writing . . . 7

6. General UI . . . 8

Part 3: Starting values (references/values.md) . . . 10

1. Fonts . . . 10

2. Type scale . . . 11

3. Spacing scale . . . 11

4. Colour structure . . . 11

5. Radius . . . 12

6. Texture (film grain) . . . 12

7. Shadows . . . 12

8. Motion . . . 13

9. Snippets . . . 13

Website optimizer

Skill metadata

name	website-optimizer
description	Build or redesign websites and web UI so they look deliberately designed instead of AI-generated. Covers typography, colour, texture, layout, copy, icons, imagery, accessibility and interaction states, plus a domain-by-domain review method for auditing an existing site. Use this skill whenever the user asks for a website, landing page, homepage, portfolio, marketing page, web app UI, React/HTML/Tailwind frontend, or says their site "looks AI-generated", "looks generic", "looks like a template", "looks like every other site", or asks to make a frontend more original, polished, distinctive or beautiful, even if they never mention AI.

Generated interfaces look alike because a model reaches for the most common choice at every decision point: the default font, the startup palette, the centred hero, three cards, "Elevate your business". No single choice is wrong. The sum of them reads as "nobody decided anything".

This skill replaces defaults with decisions. Two rules sit above everything else:

1. **Every visible choice needs a reason tied to this specific project.** "Warm near-black because the bakery's brand is about warmth" is a decision. "It was already there" is a default.
2. **Do not swap one uniform rulebook for another.** If you apply the bans below mechanically, you produce the next generic look. Break a rule when the project gives you a reason, and say what the reason is.

Workflow

1. Find the direction before writing code

Pull material from the subject, not from the genre. Answer these in a short design note (keep it in your response or a comment at the top of the stylesheet):

- **The real thing.** What physical place, object, product or person is this site about? A boxing gym in a converted mill, a sourdough bakery, a solo accountant. If the user gave no specifics, ask for them or use what you have; do not invent a business.
- **Palette source.** Sample colours from a real object connected to the brand: the shop's awning, a product photo, the packaging, a book cover, the neighbourhood's brick. Write down the object. A palette pulled from a photograph cannot drift to indigo by accident.
- **Temperature.** Pick warm or cool and hold it across every neutral.
- **Type pairing.** One display face with character, one quiet text face, or a single well-chosen family. State why it fits.
- **One signature detail.** Something specific to this project that a template would not have: a hand-drawn mark, a real map, a timetable set like a printed notice, a texture from the material the business works with.

2. Build first

Build a complete working version before polishing. Taste is easier to apply to something that exists. Use the defaults in `references/values.md` for concrete numbers (type scale, spacing, radius, grain, shadows) instead of inventing values on the fly.

3. Review by domain, one pass each

Read `references/checklist.md` and run six separate passes over the build, one per domain:

1. Accessibility
2. Layout and spacing
3. Typography
4. Colour and surfaces
5. Writing (copy)
6. General UI (components, icons, imagery, states, motion)

Separate passes find more than one general pass, because a general pass stops at the loudest problem in each area.

Rules for findings:

- **Evidence and location.** Quote the exact offending value and where it lives (`text-[15px]` in `Hero.tsx` line 40). A finding without evidence is an opinion; drop it.
- **Group by root cause.** Fifteen off-scale font sizes are one finding with fifteen locations.
- **Severity.** High: misleads the user, fabricates something, or makes the page unmistakably AI-generated. Medium: clearly hurts quality. Low: isolated polish.
- **Zero findings is a valid result.** Never pad a review to look thorough. If a domain is clean, say so.

4. Fix in impact-per-risk order

1. Font
2. Colour and surfaces
3. Interaction states
4. Layout and spacing
5. Motion
6. Generic components
7. Empty and error states
8. Copy
9. Type-scale polish

The font change is usually one line and does the most. Copy comes late because it often depends on real information from the user.

5. Report the decisions

When you hand the work back, list the main choices and the reason for each, including any rule from this skill you deliberately broke. Flag anything you could not make real (testimonials, numbers, photos) instead of filling it with invented content.

The short version of the tells

Use this as a quick scan. The checklist has the fix for each.

Type: Inter, Roboto, Arial, Open Sans or Helvetica as an unexamined choice. Arbitrary sizes off the scale. Italics for emphasis. Title Case On Buttons. Two near-identical sans faces. Orphaned last words in headings.

Colour and surface: purple, indigo or blue by default. Gradients, especially a purple-to-indigo hero. Mixed warm and cool greys. More than one accent. Pure `#000` on pure `#fff`.

Layout: three equal cards in a row. Everything centred. Every group has exactly three items. Cards wrapping everything. Pricing cards whose buttons sit at different heights. `height: 100vh`. Shadows falling in different directions. `z-index: 9999`. Radii chosen by feel.

Copy: em dashes. Hype words (elevate, seamless, unleash, supercharge, next-gen, game-changer, delve, tapestry, "in the world of"). "Learn more" and "Submit". Slogans that fit any business. Round invented numbers (99.9%, 10,000+, 24/7). Placeholder names (Acme, John Doe, Nexus). Exclamation marks and "Oops!". Fabricated testimonials, ratings, logos or awards.

Icons and images: untouched Lucide, Feather or Material sets. Rocket for launch, shield for security, lightning bolt for speed. The first stock photo anyone would think of for that industry. Emoji as feature icons.

Accessibility and behaviour: broken landmarks. Auto-moving content with no pause. No `prefers-reduced-motion`. Scroll listeners for reveal animations. Only the happy path built (no hover, focus, active, loading, empty or error states).

Honesty rule

Invented social proof is not a style problem, it is a false statement to the site's visitors. Never write testimonials, star ratings, review counts, client logos, awards, certifications, user counts or statistics that the user did not supply. Leave a clearly marked slot (`<!-- TODO: real testimonial from client -->`) or remove the section. A generated illustration is decoration; a generated five-star review on a real business's site is a lie.

Reference files

- `references/checklist.md`: every tell with its fix, grouped by the six review domains. Read before step 3.
- `references/values.md`: concrete starting values (fonts, type scale, spacing, radius formula, grain texture, shadows, motion) and code snippets. Read during step 2.

Review checklist

One section per review pass. For each item: what to look for, why it reads as generated, and what to do instead. Record findings with the exact value and its location.

Contents

1. Accessibility
2. Layout and spacing
3. Typography
4. Colour and surfaces
5. Writing
6. General UI: components, icons, imagery, states, motion

---

1. Accessibility

Generated sites skip these most reliably. They are requirements, not taste.

Landmarks. Look for `<header>` or `<footer>` nested inside `<main>`. It renders the same but breaks landmark navigation for screen reader users. `header` (banner), `main` and `footer` (contentinfo) should be siblings under `body`. One `main` per page. One `h1`, and heading levels that do not skip.

Auto-moving content. Carousels, marquees, autoplaying video, ticker text. Anything that moves on its own for more than five seconds needs a pause control (WCAG 2.2.2). Give it a real `<button>` with `aria-pressed`. For video, derive the button state from the element's own `play` and `pause` events, not from a variable you set when clicked, because browsers pause background-tab video on their own and the button will show the wrong state.

Reduced motion. Every animation and transition beyond simple colour changes goes inside `@media (prefers-reduced-motion: no-preference)` or has a reduced fallback.

Focus. Every interactive element has a visible `:focus-visible` style. Never `outline: none` without a replacement.

Contrast. Body text at least 4.5:1, large text and UI boundaries at least 3:1. Light grey text on off-white is a common failure.

Names. Icon-only buttons have `aria-label`. Images have meaningful `alt` or `alt=""` if decorative. Form fields have visible `<label>`s, not placeholder-only.

Hit areas. Tap targets at least 44 by 44 px on touch layouts.

2. Layout and spacing

Three equal cards in a row. The single most generic layout. Three to five benefits is fine as content; the equal three-column card row is the wrong container. Use a bento grid with unequal cells, an icon list, a numbered list, a zig-zag of image and text splits, or a table. If you do use three cards, use them once on the page.

Everything centred. Centred hero, centred sections, centred text blocks. Symmetry is what appears when no decision was made. Left-align body copy. Offset the hero copy against an image. Anchor imagery to an edge or the bottom of a section. Keep centring for short, single-purpose moments.

Every group has three items. Three features, three stats, three testimonials, three steps. Use the real count. Two or four or seven is fine.

Cards everywhere. Border plus shadow plus white background around every block flattens the hierarchy. Separate content with background colour alone, spacing alone, or a rule. Keep true cards for things that are actually separate objects (a product, a post) or need elevation.

Ragged card grids. In pricing tables and card rows, titles, prices, lists and buttons start at different heights. Fix: fixed-height or `min-height` header block, `min-height` on descriptions, and the card as a flex column with `margin-top: auto` on the CTA so all buttons share one baseline. Or use CSS subgrid.

Viewport height. `height: 100vh` jumps on mobile Safari as the toolbar moves. Use `min-height: 100dvh` (with `100vh` before it as a fallback).

Shadow direction. Audit every `box-shadow`. They should all imply one light source (usually above, so positive y offset, near-zero x). Mixed offsets look like two lamps.

z-index escalation. `z-index: 9999` means the stacking context is wrong. Define a small scale (base, dropdown, sticky, overlay, modal, toast) and use `isolation: isolate` on components that need their own context.

Radius by feel. Nested rounded shapes whose corners fight. Use concentric radius: inner radius = outer radius minus the gap between them (when the gap is under about 32px). A 20px container with 8px padding holds a 12px button.

Spacing off the scale. `mt-[13px]`, `gap: 22px`. Snap to the spacing scale in `values.md`. Use more space between sections than within them so grouping is obvious.

3. Typography

Default typeface. Inter, Roboto, Arial, Open Sans, Helvetica, system-ui as the brand face. They are good fonts and obvious defaults, which tells readers nobody chose. Pick a face for a reason connected to the project (see `values.md` for a starting pool). Exception: keep a default if it is the client's established brand face, and say so.

Arbitrary sizes. `text-[15px]`, `1.4rem`, `17px` scattered through the code. Each is invisible; together they make the page feel slightly off. Snap every size to the type scale and use the line height paired with that step.

Italics for emphasis. Use one weight step up instead (400 to 500 or 600). Keep italics for their real typographic jobs: titles of works, foreign terms, a deliberate display style.

Weight extremes. No thin or light weights at body size (unreadable). Black and extra-bold only for large display type, and only if the face was drawn for it.

Title Case On Every Word. Use sentence case for headings, buttons, nav and labels. For labels that need to look like labels, use a small eyebrow style: small size, letter-spacing around 0.06em, uppercase or small caps, muted colour.

Two similar sans faces. Pairing Inter with Manrope reads as a mistake. If you use two families, make them contrast: a serif or characterful display face over a quiet sans, or a sans over a mono for data.

Orphans and rags. A single word alone on the last line of a heading. Apply `text-wrap: balance` to headings and `text-wrap: pretty` to paragraphs instead of manual `<br>`s.

Line length. Body text 60 to 75 characters per line (`max-width: 65ch` or so).

Numbers. Use `font-variant-numeric: tabular-nums` in tables, prices and counters so digits align.

4. Colour and surfaces

Startup palette. Purple, violet, indigo, electric blue, with a lighter tint of the same. Models land here because it is what they saw most. Sample the palette from a real object tied to the brand and write the object down.

Gradients. Linear, radial, mesh, gradient text, gradient borders, glowing blobs behind the hero. The purple-to-indigo hero is the most recognisable generated-site marker. Remove them and use flat colour. If a gradient survives, it needs a specific reason (it depicts a real thing like a sunset photo or a material).

Flatness after removing gradients. Solve it with texture and depth, not gradients: subtle film grain (see `values.md`), real photography, layered elements that overlap section boundaries, shadows tinted with the background hue instead of grey.

Mixed temperature. Warm off-white background with cool slate grey text, or Tailwind `gray` beside `stone`. Pick one temperature and derive every neutral from it.

Competing accents. Two or more saturated colours fighting for attention. One accent per site. Everything else is neutral or a tint of the accent.

Pure black on pure white. Harsh, and a common default. Use a near-black and an off-white from your temperature.

Dark mode as inversion. Swapping white for black. Dark surfaces should be slightly lifted near-blacks with reduced-saturation accents, and elevation shown by lighter surfaces rather than shadows.

5. Writing

Em dashes. The quickest marker of machine-written copy at the moment. Rewrite with a full stop, comma, colon or parentheses.

Hype vocabulary. Elevate, seamless, unleash, supercharge, empower, revolutionise, next-gen, cutting-edge, game-changer, delve, tapestry, robust, leverage, "in today's fast-paced world", "in the world of", "take it to the next level". If a sentence needs one of these, it has nothing to say. Replace it with a concrete fact.

Generic CTAs. "Learn more", "Get started", "Submit", "Click here". Name the outcome: "Book a free first class", "See this week's timetable", "Send message". Three "Learn more" links on one page each need to say more about what.

Interchangeable slogans. "Quality you can trust", "Train hard, fight smart", "Your success is our mission". Test: could this headline hang outside any business of the same type? If yes, rewrite from specifics: the place, the offering, the customer. "Learn to box in Ancoats" beats any slogan.

Round invented numbers. 99.9% uptime, 10,000+ customers, 24/7 support, 50% faster. Models fill slots with these. Real numbers are lumpy ("since 2011", "4.9 miles from the station", "about 300 loaves a week"). Use only numbers the user gave you.

Placeholder names. Acme Corp, John Doe, Jane Smith, Nexus, SmartFlow, Lorem ipsum in a shipped page. Use the real names or mark the slot clearly.

Cheerful errors. "Oops! Something went wrong!" Error copy states what happened and what to do: "Couldn't connect. Check your connection and try again." No exclamation marks in interface copy.

Fabricated proof. Testimonials, star ratings, review counts, client logo walls, awards, press mentions, credentials. If the user has none, there is no section. See the honesty rule in SKILL.md.

Feature-list voice. Every section opening with a bolded noun phrase followed by a colon and a benefit. Vary structure; write like a person describing their own business.

6. General UI

Default icon sets used raw. Lucide, Feather, Heroicons, Material at default stroke and size. Acceptable as a base, but adjust stroke width to match the type weight, size them to the text they sit beside, and choose less obvious metaphors (Lock rather than Shield for security, AlertTriangle for warnings, Fingerprint for integrity). Avoid the rocket, lightning bolt, sparkles and brain icons. No emoji as feature icons.

First-thought imagery. Handshake for partnership, laptop on a desk for software, smiling team at a whiteboard. Use the user's real photos first. Then real stock chosen for specificity. If generating images, fix a style vocabulary (medium, lighting, palette, lens, composition) and hold it constant while changing the subject so the set looks art-directed.

Over-cleaned artwork. Redrawing a hand-made or generated mark into perfect geometric SVG often makes it worse. Slight irregularity is what makes a mark read as an object rather than a template. Keep it unless the user asks otherwise.

Missing states. Generated UIs ship the happy path only. Every interactive element needs hover, active, focus-visible and disabled. Every data view needs loading, empty and error states, each with useful copy. Forms need inline validation messages.

Scroll-listener reveals. `window.addEventListener('scroll', ...)` for fade-in effects causes constant layout work and janky mobile scrolling. Use `IntersectionObserver`, or CSS scroll-driven animations where supported.

Motion everywhere. Every section fading up on scroll is its own tell. Animate where motion explains something (a state change, a reveal the user triggered). Keep durations short (150 to 300ms for UI) and ease-out for entrances.

Generic component shapes. The pill badge above the hero headline ("New: AI-powered..."), the stat row of three big numbers under the hero, the floating-blob background, glassmorphism cards. Remove unless they carry real information the user supplied.

Hero for the sake of a hero. A full-viewport hero with a headline and two buttons before any content. Consider leading with the actual product, menu, timetable or work.

Starting values

Use these instead of inventing numbers. They are starting points; change any of them for a project-specific reason and record the reason.

Print note: long lines in the code blocks below are wrapped to fit the page. In values.md each of them is a single line, including the grain data URI.

Contents

1. Fonts
2. Type scale
3. Spacing scale
4. Colour structure
5. Radius
6. Texture (film grain)
7. Shadows
8. Motion
9. Snippets

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1. Fonts

Avoid as an unexamined brand face: Inter, Roboto, Arial, Open Sans, Helvetica, system-ui. Also avoid reaching for the same replacement every time; rotate and match the project.

Starting pool (all on Google Fonts or free to self-host):

- **Text and UI sans:** Manrope, Geist, Instrument Sans, Figtree, Public Sans, Schibsted Grotesk, Hanken Grotesk, DM Sans, Onest
- **Display sans with character:** Bricolage Grotesque, Space Grotesk, Syne, Familjen Grotesk, Unbounded (sparingly)
- **Serif, display or text:** Fraunces, Newsreader, Instrument Serif, Source Serif 4, Literata, Young Serif, EB Garamond
- **Mono for data, code, labels:** Geist Mono, JetBrains Mono, IBM Plex Mono, DM Mono

Pairing patterns that work:

- Characterful serif display (Fraunces, Instrument Serif) over a quiet sans text face (Manrope, Public Sans)
- One sans family for everything, with contrast from size and weight only
- Sans text face plus mono for numbers, labels and metadata

Weights: 400 body, 500 or 600 for emphasis and UI labels, 600 to 700 for headings. Nothing above 700 unless the display face was drawn for it. Nothing below 400 at body sizes.

Load with `font-display: swap` and give a real fallback stack, for example `"Manrope", ui-sans-serif, system-ui, sans-serif`.

2. Type scale

Ratio about 1.25 (major third). Each step has its own line height and tracking. Use only these steps.

Token	Size	Line height	Tracking	Use
xs	0.75rem / 12px	1.5	0.02em	Captions, legal
sm	0.875rem / 14px	1.5	0	UI labels, meta
base	1rem / 16px	1.6	0	Body
lg	1.25rem / 20px	1.5	0	Lead paragraphs
xl	1.5625rem / 25px	1.3	-0.01em	h3
2xl	1.953rem / 31px	1.2	-0.015em	h2
3xl	2.441rem / 39px	1.15	-0.02em	h1 on inner pages
4xl	3.052rem / 49px	1.1	-0.025em	Hero h1 (mobile can drop one step)
5xl	3.815rem / 61px	1.05	-0.03em	Hero h1, large screens only

Fluid headings are fine with `clamp()`, but the min and max must both be steps on this scale.

Eyebrow label: sm or xs, weight 500 to 600, `letter-spacing: 0.06em`, uppercase or `font-variant-caps: all-small-caps`, muted colour.

3. Spacing scale

4px base: 4, 8, 12, 16, 24, 32, 48, 64, 96, 128. Inside components use 4 to 24. Between related blocks use 24 to 48. Between page sections use 64 to 128. Sections should be clearly further apart than the things inside them.

Content width: body text `max-width: 65ch`. Page container around 1120 to 1280px with 16 to 24px side padding on mobile.

4. Colour structure

Build tokens from the sampled object, then fill these roles:

- `--bg`: off-white or near-black in your chosen temperature (warm example `#f7f3ec`, cool example `#f3f5f7`)
- `--surface`: one step lifted from `--bg`
- `--text`: near-black in the same temperature (warm example `#1a1310`, cool example `#11161c`)
- `--text-muted`: text colour at reduced lightness contrast, still 4.5:1 on `--bg`
- `--border`: low-contrast line from the same family
- `--accent`: the one saturated colour, sampled from the object

- `--accent-ink`: text colour that passes contrast on `--accent`

Neutrals for a warm palette come from Tailwind `stone` or custom warm greys; for a cool palette from `slate` or `zinc`. Never mix the two families.

Check every text and background pair for contrast before finishing.

5. Radius

Pick one base radius for the project (for example 4px for sharp and editorial, 12px for soft and friendly) and derive the rest.

Concentric rule for nested shapes, when the gap is under about 32px:

```
inner radius = outer radius - gap
```

Example: input container 20px radius, 8px padding, button inside at 12px. With a larger gap the shapes no longer read as concentric and can be chosen independently.

6. Texture (film grain)

Adds depth to flat colour without a gradient. Apply as a pseudo-element overlay.

```
:root {
  --grain: url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' width='200'
    height='200'%3E%3Cfilter id='n'%3E%3CfeTurbulence type='fractalNoise' baseFrequency='0.85'
    numOctaves='3' stitchTiles='stitch'/%3E%3C/filter%3E%3Crect width='100%25' height='100%25'
    filter='url(%23n)'/%3E%3C/svg%3E");
}

.grain { position: relative; isolation: isolate; }
.grain::after {
  content: "";
  position: absolute;
  inset: 0;
  background-image: var(--grain);
  opacity: 0.06;            /* light surfaces */
  pointer-events: none;
  z-index: 1;
}
.grain--colour::after {
  opacity: 0.16;            /* saturated or dark panels */
  mix-blend-mode: overlay;
}
```

Keep content above the overlay (`position: relative; z-index: 2` on children) or place the overlay behind content with `z-index: -1`.

7. Shadows

One light source, from above. Tint the shadow with the background hue rather than using neutral grey.

```
--shadow-sm: 0 1px 2px rgb(40 25 15 / 0.06), 0 1px 1px rgb(40 25 15 / 0.04);
--shadow-md: 0 4px 12px rgb(40 25 15 / 0.08), 0 1px 3px rgb(40 25 15 / 0.06);
--shadow-lg: 0 16px 40px rgb(40 25 15 / 0.12), 0 4px 12px rgb(40 25 15 / 0.06);
```

(Warm tint shown. For a cool palette use something like `rgb(15 25 40 / ...)`.) All x offsets 0.

8. Motion

- UI feedback (hover, press): 120 to 180ms, `ease-out`
- Entrances and reveals: 200 to 320ms, `cubic-bezier(0.2, 0.8, 0.2, 1)`
- Animate `transform` and `opacity` only
- Wrap all non-essential motion:

```
@media (prefers-reduced-motion: no-preference) {
  .reveal { transition: opacity 280ms cubic-bezier(.2,.8,.2,1), transform 280ms
    cubic-bezier(.2,.8,.2,1); }
}
```

9. Snippets

Balanced headings, no orphans

```
h1, h2, h3 { text-wrap: balance; }
p { text-wrap: pretty; }
```

Full-height section

```
.hero { min-height: 100vh; min-height: 100dvh; }
```

Aligned pricing cards

```
.plans { display: grid; grid-template-columns: repeat(auto-fit, minmax(16rem, 1fr)); gap:
  24px; }
.plan { display: flex; flex-direction: column; }
.plan__head { min-height: 7rem; }      /* title + price block */
.plan__desc { min-height: 3.2em; }
.plan__cta  { margin-top: auto; }      /* every button on one line */
```

Reveal on scroll without scroll listeners

```
const io = new IntersectionObserver((entries) => {
  for (const e of entries) {
    if (e.isIntersecting) { e.target.classList.add('is-visible'); io.unobserve(e.target); }
  }
}, { rootMargin: '0px 0px -10% 0px' });
document.querySelectorAll('.reveal').forEach((el) => io.observe(el));
```

Pause button for autoplaying video (state follows the video)

```
const video = document.querySelector('#hero-video');
const btn = document.querySelector('#hero-pause');
const sync = () => {
  const playing = !video.paused;
  btn.setAttribute('aria-pressed', String(!playing));
  btn.textContent = playing ? 'Pause video' : 'Play video';
};
video.addEventListener('play', sync);
video.addEventListener('pause', sync);
btn.addEventListener('click', () => (video.paused ? video.play() : video.pause()));
sync();
```

z-index scale

```
:root { --z-base: 0; --z-dropdown: 10; --z-sticky: 20; --z-overlay: 30; --z-modal: 40;
  --z-toast: 50; }
```